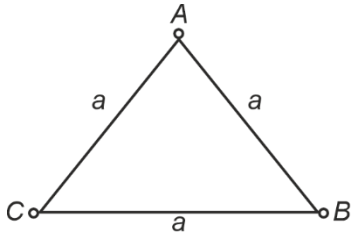
PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

26.



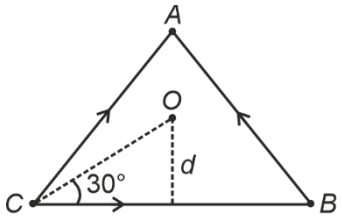
Three equal masses m are kept at vertices (A, B, C) of an equilateral triangle of side a in free space. At $t = 0$, they are given an initial velocity $\vec{V_A} = V_0\,\vec{AC}$, $\vec{V_B} = V_0\,\vec{BA}$ and $\vec{V_C} = V_0\,\vec{CB}$. Here, $\vec{AC}$, $\vec{CB}$ and $\vec{BA}$ are unit vectors along the edges of the triangle. If the three masses interact gravitationally, then the magnitude of the net angular momentum of the system at the point of collision is :

(1) $\frac{1}{2}amV_0$ (2) $\frac{3}{2}amV_0$

(3) $\frac{\sqrt{3}}{2}amV_0$ (4) $3amV_0$

Answer (3)

Sol.



$d = \frac{a}{2\sqrt{3}}$

Angular momentum of one mass about point O

$L = mvd$

$= mv_0 \cdot \frac{a}{2\sqrt{3}}$

Net angular momentum about point O

$L_{net} = 32$

$= \frac{\sqrt{3}mv_0a}{2}$

27. Two bodies A and B of equal mass are suspended from two massless springs of spring constant k_1 and k_2, respectively. If the bodies oscillate vertically such that their amplitudes are equal, the ratio of the maximum velocity of A to the maximum velocity of B is

(1) $\sqrt{\frac{k_1}{k_2}}$ (2) $\frac{k_1}{k_2}$

(3) $\frac{k_2}{k_1}$ (4) $\sqrt{\frac{k_2}{k_1}}$

Answer (1)

Sol. Here $\omega = \sqrt{\frac{k}{m}}$

and maximum velocity $V = A\omega = A\sqrt{\frac{k}{m}}$

So, $\frac{V_A}{V_B} = \sqrt{\frac{k_1}{k_2}}$

28. A point charge causes an electric flux of -2×10^4 Nm^2C^{-1} to pass through a spherical Gaussian surface of 8.0 cm radius, centred on the charge. The value of the point charge is :

(Given $\in_0 = 8.85 \times 10^{-12}$ $C^2N^{-1}m^{-2}$)

(1) 17.7×10^{-8} C (2) -15.7×10^{-8} C

(3) 15.7×10^{-8} C (4) -17.7×10^{-8} C

Answer (4)



Sol. $\text{Flux}(\phi) = \frac{\theta_{inc}}{\epsilon_0}$

$\theta_{inc} = \epsilon_0 \phi$

$= -17.7 \times 10^{-8}$ C

29. In an experiment with photoelectric effect, the stopping potential,

(1) decreases with increase in the intensity of the incident light

(2) is $\left(\frac{1}{e}\right)$ times the maximum kinetic energy of the emitted photoelectrons

(3) increases with increase in the wavelength of the incident light

(4) increases with increase in the intensity of the incident light

Answer (2)

Sol. From Einstein photoelectric equation

$\frac{hc}{\lambda} = \phi + eV_S$

Maximum K.E. = $(K)_{max} = eVs$

So, $V_S = \frac{(K)_{max}}{e}$

30. The number of spectral lines emitted by atomic hydrogen that is in the 4th energy level, is

(1) 6 (2) 0

(3) 3 (4) 1

Answer (1)

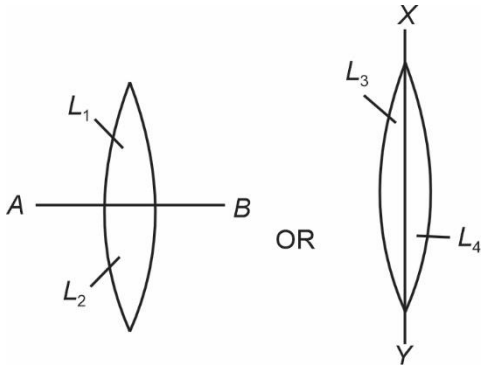
Sol. Number of spectral line $(N) = \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$

Here $n_2 = 4$

$n = 1$

So, $N = 6$

31. Two identical symmetric double convex lenses of focal length f are cut into two equal parts L_1, L_2 by AB plane and L_3, L_4 by XY plane as shown in figure respectively. The ratio of focal lengths of lenses L_1 and L_3 is

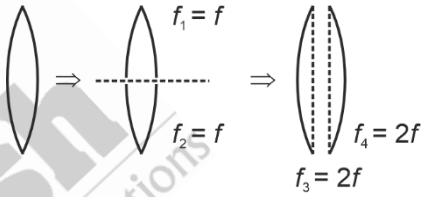


(1) 2 : 1 (2) 1 : 2

(3) 1 : 1 (4) 1 : 4

Answer (2)

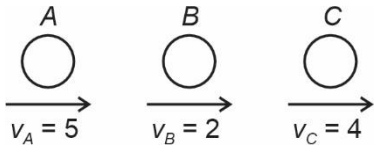
Sol.



so $\frac{f_1}{f_3} = 1:2$

32. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) :



Three identical spheres of same mass undergo one dimensional motion as shown in figure with initial velocities v_A = 5 m/s, v_B = 2 m/s, v_C = 4 m/s. If we wait sufficiently long for elastic collision to happen, then v_A = 4 m/s, v_B = 2 m/s, v_C = 5 m/s will be the final velocities.

Reason (R) : In an elastic collision between identical masses, two objects exchange their velocities.



In the light of the above statements, choose the **correct** answer from the options given below.

(1) **(A)** is true but **(R)** is false

(2) **(A)** is false but **(R)** is true

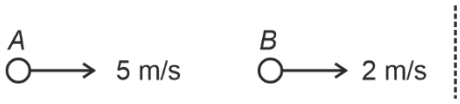
(3) Both **(A)** and **(R)** are true but **(R)** is **NOT** the correct explanation of **(A)**

(4) Both (A) and **(R)** are true and **(R)** is the correct explanation of **(A)**

Answer (2)

Sol. For A and B.

Before collision



After collision



For B and C

Before collision



After collision



Final velocity

$V_A = 2$ m/s

$V_B = 4$ m/s

$V_C = 5$ m/s

$\Rightarrow$ Velocity exchange for two identical mass in elastic collision.

33. Given below are two statements : One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : With the increase in the pressure of an ideal gas, the volume falls off more rapidly in an isothermal process in comparison to the adiabatic process.

Reason (R) : In isothermal process, PV = constant, while in adiabatic process PV^{γ} = constant. Here γ is the ratio of specific heats, P is the pressure and V is the volume of the ideal gas.

In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both (A) and **(R)** are true and **(R)** is the correct explanation of **(A)**

(2) **(A)** is false but **(R)** is true

(3) Both **(A)** and **(R)** are true but **(R)** is **NOT** the correct explanation of **(A)**

(4) **(A)** is true but **(R)** is false

Answer (1)

Sol. For isothermal process

PV = constant

$\Rightarrow P\Delta V + V\Delta P = 0$

$\Rightarrow \dfrac{\Delta V}{V} = \dfrac{-\Delta P}{P}$

$\Rightarrow \Delta V = -\left(\dfrac{V}{P}\right)\Delta P$

For adiabetic PV^{γ} = constant

$\Rightarrow P^{\gamma}V^{\gamma-1}dV + V^{\gamma}dP = 0$

$\Rightarrow dV = \dfrac{-V}{\gamma P}(dP)$

Magnitude of $\left|\dfrac{V}{P}\right|$ is greater than $\left|\dfrac{V}{\gamma P}\right|$

So, volume falls more rapidly in isothermal.

2nd statement is the process description of the isothermal and adiabatic



34. The difference of temperature in a material can convert heat energy into electrical energy. To harvest the heat energy, the material should have

(1) Low thermal conductivity and high electrical conductivity

(2) High thermal conductivity and high electrical conductivity

(3) Low thermal conductivity and low electrical conductivity

(4) High thermal conductivity and low electrical conductivity

Answer (1)

Sol. Material should have low thermal conductivity and high electrical conductivity.

35. A sand dropper drops sand of mass $m(t)$ on a conveyer belt at a rate proportional to the square root of speed (v) of the belt, i.e. $\frac{dm}{dt} \propto \sqrt{v}$. If P is the power delivered to run the belt at constant speed then which of the following relationship is true ?

(1) $P \propto v$ (2) $P \propto \sqrt{v}$

(3) $P^2 \propto v^5$ (4) $P^2 \propto v^3$

Answer (3)

Sol. Power = $\vec{F}.\vec{V}$

and $F = \frac{dp}{dt}$ = Rate of change of linear momentum

$F = V\cdot\frac{dm}{dt} = K_1V^{\frac{3}{2}}$, K is constant

Power (P) = $\left(KV^{\frac{3}{2}}\right)\cdot(V)$

$= KV^{\frac{5}{2}}$

So, $P^2 \propto V^5$

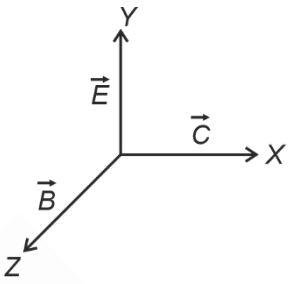
36. A plane electromagnetic wave propagates along the + x direction in free space. The components of the electric field, $\vec{E}$ and magnetic field, $\vec{B}$ vectors associated with the wave in Cartesian frame are

(1) E_x, B_y (2) E_y, B_z

(3) E_y, B_x (4) E_x, B_z

Answer (2)

Sol.



The direction of wave propagation $\rightarrow \vec{E}\times\vec{B}$

37 Match **List** - I with **List - II**.

	List - I		**List - II**
(A)	Magnetic induction	(I)	Ampere meter^2
(B)	Magnetic intensity	(II)	Weber
(C)	Magnetic flux	(III)	Gauss
(D)	Magnetic moment	(IV)	Ampere meter

Choose the **correct** answer from the options given below :

(1) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(2) (A)-(III), (B)-(IV), (C)- (I), (D)-(II)

(3) (A)-(III), (B)-(IV), (C)- (II), (D)-(I)

(4) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

Answer (3)



Sol. (A) Magnetic induction → Gauss (III)

(B) Magnetic intensity → Ampere/meter (IV) → $\mu = \frac{B}{\mu}$

(C) Magnetic flux → weber (II) → $\phi = \vec{B} \cdot \vec{A}$

(D) Magnetic moment → Ampere-meter^2 (I) → $\vec{M} = I\vec{A}$

38. Match **List** - I with **List - II**.

	List - I		**List - II**
(A)	Young's Modulus	(I)	$ML^{-1}T^{-1}$
(B)	Torque	(II)	$ML^{-1}T^{-2}$
(C)	Coefficient of Viscosity	(III)	$M^{-1}L^{3}T^{-2}$
(D)	Gravitational Constant	(IV)	$ML^{2}T^{-2}$

Choose the **correct** answer from the options given below :

(1) (A)-(I), (B)- (III), (C)-(II), (D)-(IV)

(2) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

(3) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

(4) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

Answer (3)

Sol. (A) Young's Modulus (Y) $= Y = \frac{F}{A\left(\frac{\Delta l}{l}\right)}$

$[Y] = [ML^{-1}T^{-2}]$ …(II)

(B) Torque $(\xi^{-1}) = \vec{r} \times \vec{F}$

$[\xi] = [ML^{2}T^{-2}]$ …(IV)

(C) Coefficient of Viscosity (η)

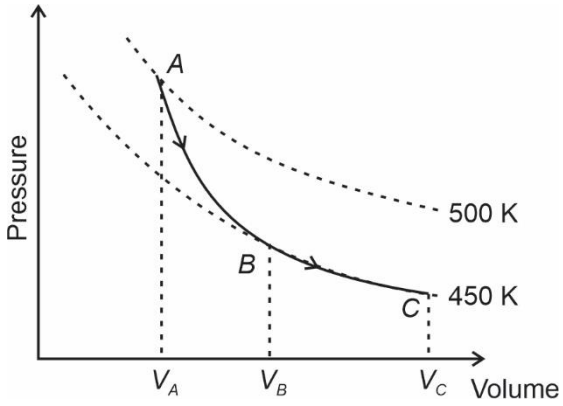
$F = \eta A \frac{dV}{dt}$

$[\eta] = [ML^{-1}T^{-1}]$ …(I)

(D) Gravitational Constant $[G] = \frac{F \cdot d^2}{m_1 m_2}$

$[G] = [M^{-1}L^{3}T^{-2}]$ …(III)

39.



A poly-atomic molecule ($C_y = 3R$, $C_p = 4R$, where R is gas constant) goes from phase space point $A(P_A = 10^5$ Pa, $V_A = 4 \times 10^{-6}$ m^3) to point $B(P_B = 5 \times 10^4$ Pa, $V_B = 6 \times 10^{-6}$ m^3) to point $C(P_C = 10^4$ Pa, $V_C = 8 \times 10^{-6}$ m^3). A to B is an adiabatic path and B to C is an isothermal path. The net heat absorbed per unit mole by the system is :

(1) 500R ln2

(2) 500R(ln3 + ln4)

(3) 400R ln4

(4) 450R(ln4 –ln3)

Answer (4)

Sol. For process $A \rightarrow B$

$(\Delta Q)_{AB} = 0$ (adiabatic)

For process $B \rightarrow C$

$(\Delta Q)_{BC} = (\Delta W)_{BC} = nRT \ln\left(\frac{V_C}{V_B}\right)$

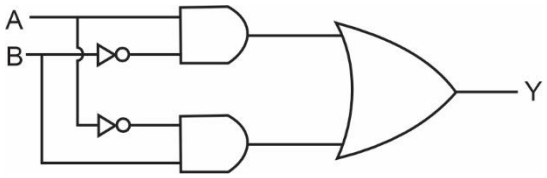
$= 450R\,[\ln(4) - \ln(3)]$

$(\Delta Q)_{net} = (\Delta Q)_{AB} + (\Delta Q)_{BC}$

$= 450R\,[\ln(4) - \ln(3)]$



40. The truth table for the circuit given below is :



(1)

A	B	Y
0	0	0
1	0	1
0	1	0
1	1	0

(2)

A	B	Y
0	0	0
1	1	1
1	0	1
0	1	1

(3)

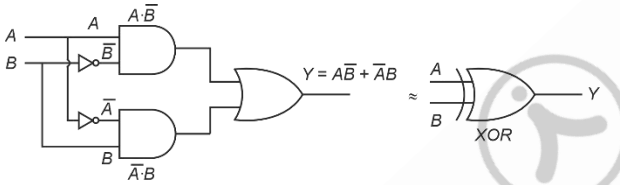
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(4)

A	B	Y
0	0	0
1	0	0
1	1	0
0	1	1

Answer (3)

Sol.



41. A cup of coffee cools from 90°C to 80°C in t minutes when the room temperature is 20°C. The time taken by the similar cup of coffee to cool from 80°C to 60°C at the same room temperature is :

(1) $\frac{13}{5}t$ (2) $\frac{10}{13}t$

(3) $\frac{13}{10}t$ (4) $\frac{5}{13}t$

Answer (1)

Sol. From Newton law

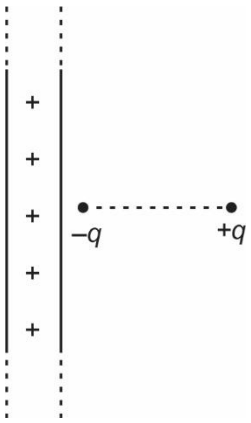
$$\frac{90-80}{t} = c\left(\frac{90+80}{2} - 20\right) \quad \ldots(i)$$

and $$\frac{80-60}{t_1} = c\left(\frac{80+60}{2} - 20\right) \quad \ldots(ii)$$

From (i) and (ii)

$$t_1 = \frac{13}{5}t$$

42. An electric dipole is placed at a distance of 2 cm from an infinite plane sheet having positive charge density σ_0. Choose the correct option from the following.



(1) Torque on dipole is zero and net force acts towards the sheet.

(2) Potential energy of dipole is minimum and torque is zero.

(3) Potential energy and torque both are maximum.

(4) Torque on dipole is zero and net force is directed away from the sheet.

Answer (2)

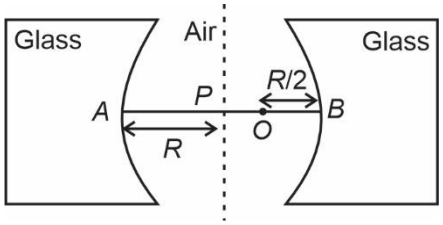
Sol. Electric field due to sheet $E = \frac{\sigma}{2\varepsilon_0}$

and torque on dipole $\vec{\tau} = \vec{P} \times \vec{E}$

here $\vec{\tau} = 0$

and $U = -\vec{P}\cdot\vec{E}$ $\rightarrow$ should be minimum



43. 

Two concave refracting surfaces of equal radii of curvature and refractive index 1.5 face each other in air as shown in figure. A point object O is placed midway, between P and B. The separation between the images of O, formed by each refracting surface is :

(1) 0.114 R (2) 0.411 R

(3) 0.214 R (4) 0.124 R

Answer (1)

Sol. For glass B

$$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$$

$$\frac{1.5}{V_B}+\frac{1}{\left(\frac{R}{2}\right)}=\frac{0.5}{-R}$$

$V_B = -0.6\ R$

For glass A

$$\frac{1.5}{V_A}+\frac{2}{3R}=\frac{0.5}{-R}$$

$$V_A=-\frac{9}{7}R$$

Distance between images

$$= 2R-\left(\frac{9}{7}R+0.6R\right)$$

$$= 0.114\ R$$

44. A convex lens made of glass (refractive index = 1.5) has focal length 24 cm in air. When it is totally immersed in water (refractive index = 1.33), its focal length changes to

(1) 24 cm (2) 48 cm

(3) 72 cm (4) 96 cm

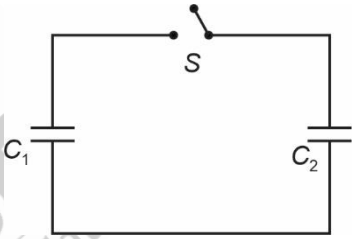
Answer (4)

Sol. $\frac{1}{24}=(1.5-1)\left(\frac{1}{R}+\frac{1}{R}\right)=\frac{1}{R}$, $R = 24$ cm

Again $\frac{1}{f'}=\left(\frac{1.5}{1.33}-1\right)\left(\frac{2}{R}\right)$

On solving, $f' = 96$ cm

45. A capacitor, $C_1 = 6\ \mu F$ is charged to a potential difference of $V_0 = 5V$ using a 5V battery. The battery is removed and another capacitor, $C_2 = 12\ \mu F$ is inserted in place of the battery. When the switch 'S' is closed, the charge flows between the capacitors for some time until equilibrium condition is reached. What are the charges (q_1 and q_2) on the capacitors C_1 and C_2 when equilibrium condition is reached.



(1) $q_1 = 15\ \mu C$, $q_2 = 30\ \mu C$

(2) $q_1 = 10\ \mu C$, $q_2 = 20\ \mu C$

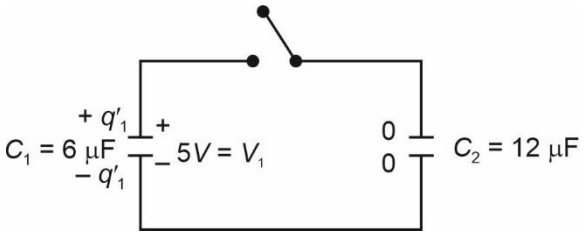
(3) $q_1 = 20\ \mu C$, $q_2 = 10\ \mu C$

(4) $q_1 = 30\ \mu C$, $q_2 = 15\ \mu C$

Answer (2)

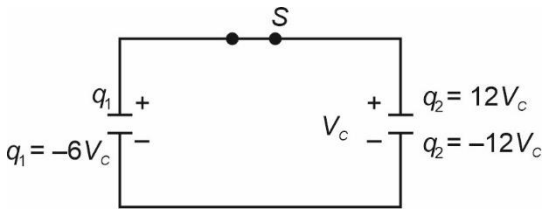
Sol. at $t = 0$

$q'_1 = C_V = 30\ \mu C$





at $t = t$



$6V_C + 12\ V_C = 30 + 0$

$V_C = \frac{5}{3} V$

$q_1 = \frac{6 \times 5}{3} = 10\ \mu C$

$q_2 = \frac{12 \times 5}{3} = 20\ \mu C$

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

46. A physical quantity Q is related to four observables a, b, c, d as follows :

$$Q = \frac{ab^4}{cd}$$

where, a = (60 ± 3) Pa; b = (20 ± 0.1) m; c = (40 ± 0.2) Nsm^{-2} and d = (50 ± 0.1) m, then the percentage error in Q is $\frac{x}{1000}$, where x = _____.

Answer (7700)

Sol. Here, $\frac{\Delta Q}{Q} \times 100 = \left[\frac{\Delta a}{a} + 4\frac{\Delta b}{b} + \frac{\Delta c}{c} + \frac{\Delta d}{d}\right] \times 100$

$$= \left[\frac{3}{60} + \frac{0.4}{20} + \frac{0.2}{40} + \frac{0.1}{50}\right]$$

$$\frac{x}{1000} = \left[\frac{3}{60} + \frac{0.4}{20} + \frac{0.2}{40} + \frac{0.1}{50}\right] \times 100$$

$x = 7700$

47. Two planets, A and B are orbiting a common star in circular orbits of radii R_A and R_B, respectively, with $R_B = 2R_A$. The planet B is $4\sqrt{2}$ times more massive than planet A. The ratio $\left(\frac{L_B}{L_A}\right)$ of angular momentum (L_B) of planet B to that of planet A(L_A) is closest to integer _____.

Answer (8)

Sol. Angular momentum, $L = mV_0R = m\sqrt{GMR}$

where M is the mass of star

$$\frac{L_B}{L_A} = \frac{m_B}{m_A}\sqrt{\frac{R_B}{R_A}} = 8$$

48. Two cars P and Q are moving on a road in the same direction. Acceleration of car P increases linearly with time whereas car Q moves with a constant acceleration. Both cars cross each other at time t = 0, for the first time. The maximum possible number of crossing(s) (including the crossing at t = 0) is _____.

Answer (3)

Sol. For first car $P \rightarrow$ acceleration (a_P) = ct, c is constant

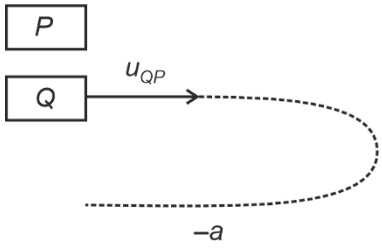
For second car $Q \rightarrow$ acceleration (a_Q) = a, a is constant

Let analyze the problem in two case

Case-I

$\Rightarrow$ U_{QP} and a_{QP} in same direction

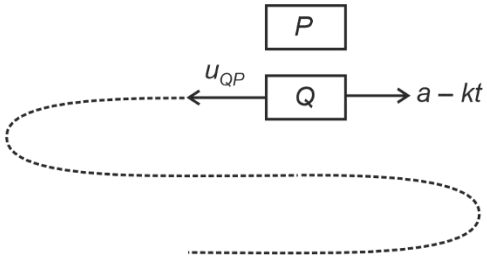




They cross 2 times in this case

Case-II

U_{PQ} and a_{QP} in opposite directions



They cross 3 time in this case

49. The magnetic field inside a 200 turns solenoid of radius 10 cm is 2.9×10^{-4} Tesla. If the solenoid carries a current of 0.29 A, then the length of the solenoid is _____ π cm.

Answer (8)

Sol. Magnetic field inside a solenoid

$$B = \mu_0 i\left(\frac{N}{L}\right)$$

$$L = \frac{\mu_0 Ni}{B} = \frac{4\pi \times 10^{-7} \times 200 \times 0.29}{2.9 \times 10^{-4}}$$

$= 8\pi$ cm

50. A parallel plate capacitor consisting of two circular plates of radius 10 cm is being charged by a constant current of 0.15 A. If the rate of change of potential difference between the plates is 7×10^8 V/s then the integer value of the distance between the parallel plates is _____ μm.

$$\left(\text{Take, } \varepsilon_0 = 9 \times 10^{-12}\,\frac{\text{F}}{\text{m}},\ \pi = \frac{22}{7}\right)$$

Answer (1320)

Sol. $Q = cV$

$$V = \frac{Q}{c} = \frac{it}{\left(\frac{\varepsilon_0 A}{d}\right)}$$

$$d = \frac{\varepsilon_0 \pi r^2}{i}\left(\frac{v}{t}\right)$$

Putting values

$$d = \frac{9 \times 10^{-12} \times \frac{22}{7} \times (0.1)^2}{0.15} \times (7 \times 10^8)$$

$= 1320$ μm



CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

51. Total number of sigma (σ) _____ and pi(π) _____ bonds respectively present in hex-1-en-4-yne are :

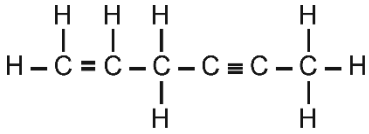
(1) 3 and 13

(2) 11 and 3

(3) 14 and 3

(4) 13 and 3

Answer (4)

Sol. 

σ bond = 13, π bond = 3

52. Drug X becomes ineffective after 50% decomposition. The original concentration of drug in a bottle was 16 mg/mL, which becomes 4 mg/mL, in 12 months. The expiry time of the drug in months is _____

Assume that decomposition of drug follows first order kinetics

(1) 6

(2) 12

(3) 3

(4) 2

Answer (1)

Sol. Drug X $\xrightarrow{\text{1st order}}$ Product

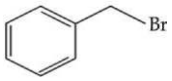
Initial concentration of drug = 16 mg/mL

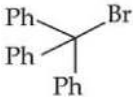
Concentration of drug after 12 months = 4 mg/mL

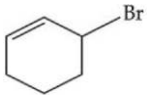
Half life of drug = 6 months

Drug become ineffective after 50% decomposition. The expiry time of drug = 6 months

53. Which among the following halides will generate the most stable carbocation in the nucleophilic substitution reaction?

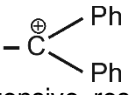
(1) 

(2) 

(3) 

(4) 

Answer (2)

Sol.  is most stable carbocation due to extensive resonance stabilization from the three phenyl groups.

54. 0.1 M solution of KI reacts with excess of H_2SO_4 and KIO_3 solutions. According to equation

$5I^- + IO_3^- + 6H^+ \rightarrow 3I_2 + 3H_2O$

Identify the **correct** statements :

(A) 200 mL of KI solution reacts with 0.004 mol of KIO_3

(B) 200 mL of KI solution reacts with 0.006 mol of H_2SO_4

(C) 0.5 L of KI solution produced 0.005 mol of I_2

(D) Equivalent weight of KIO_3 is equal to $\left(\frac{\text{Molecular weight}}{5}\right)$

Choose the **correct** answer from the options given below :

(1) (A) and (D) only (2) (C) and (D) only

(3) (A) and (B) only (4) (B) and (C) only

Answer (1)

Sol. $E_{KIO_3} = \frac{\text{Molecular weight}}{n_f}$

$n_f = 5$

$E_{KIO_3} = \frac{\text{Molecular weight}}{5}$

(D) is correct

meq of KI = $0.1 \times 200 = 20$

meq of $KIO_3 = 4 \times 5 = 20$

(A) is correct

